



The SGTE collection of binary datasets

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ABSTRACT

SGTE has collected (and also produced) binary assessments since the 1980's. This was documented with about 450 binary systems in five volumes of the Landolt-Börnstein series. These volumes were accompanied by a database with which the phase diagrams could be calculated, but the database itself was encrypted. It is now time to make the SGTE collection of binary datasets publicly available. The collection now contains 2773 datasets from 1238 different binary systems. By providing this collection in open form it is hoped that Calphad modelling of, in particular, ternary and higher order systems can be further promoted.

1. Introduction

Calphad models of binary systems (called datasets here) form the basis for further modelling of ternary (and sometimes higher order) systems and the construction of Calphad databases of any size. For these purposes a high quality of the binary models is essential, not only in terms of their ability to reproduce the intended binary phase diagrams, but also, and more importantly, a good representation of the thermodynamic properties (also outside the stable ranges of the involved phases). However, it is very difficult (or even impossible) to determine the quality of a binary dataset from the information within the binary system alone. The quality (usefulness) of a binary dataset only becomes apparent after it has been used to model ternary (and higher order) systems, and there are several examples in the literature where a key ternary system could only be modelled after changes in one or more of its binary subsystems. Therefore, the construction of thermodynamic databases is a highly iterative process where problems found in higher order systems can often only be solved by identifying the binary system(s) causing the problems. In order to construct databases, it is, thus, important to have access to as many existing binary datasets as possible in a directly useable format. Then it is relatively easy to test available datasets and to make modifications when needed. A large number of binary assessments have been published in the open literature, but it is often not easy to produce a correct database file from the information (parameter list) given in the paper. An increasing number of publications additionally provide supplementary database files, improving the situation. It is the purpose of this paper to provide as many published binary datasets as possible in a standardized format for the benefit of the Calphad community. Currently 2773 datasets from 1238 different systems are included. Considering 80 elements there are 3160 possible binary systems, i.e. there are still many binary systems for which there is no published assessment. Included systems are indicated in an Excel

table provided as supplemental material.

Thermodynamic descriptions of binary (and higher order) systems have been collected by SGTE since the mid/late 1980's to build the SGTE solution database, but no systematic collection was started at that time. In the late 1990's a more systematic collection of binary systems was started with the purpose to form the basis for five volumes of calculated binary phase diagrams in the Landolt-Börnstein series [1]. This collection contains datasets for about 450 different binary systems. For each system the best available dataset was selected and included in the collection. In contrast to this, the present collection includes all available datasets, including the mentioned 450 datasets, from which the intended (within some limits) phase diagram could be calculated. There is no criterion for quality (or lack thereof). Datasets which could not be modified to produce the intended phase diagram are not included. Some datasets represent only minor modifications of earlier datasets. For practical reasons these are usually included as separate datasets, even when the differences are not visible in the phase diagram. Therefore, the total number of datasets is quite a bit larger than the total number of actual assessments. The datasets are written in tdb (thermodynamic database file) format, which is the most commonly used format. This format is not well documented and different software use different flavours. The present datasets have been tested using Thermo-Calc software [2], which is relatively strict regarding formatting. The version of Thermo-Calc used varies from version R for some of the older datasets to version 2024b for the most recent datasets. All datasets have been modified to be accepted by the more recent Gibbs-Energy-System (GES 6) in Thermo-Calc, but some issues can still remain with older datasets. Some modification might still be necessary for use with other software. Due to the selected format and the software used, it was not possible to include binary datasets where the modified quasi-chemical model [3] is used to model the liquid phase. Most datasets belong to what is now called 2nd generation Calphad and are based on version 5.0 of the SGTE

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unary database for the element data, originating from the publication by Dinsdale [4]. Older datasets have only been included when acceptable results were achieved using SGTE element data. A small number of 3rd generation [5] datasets are also included, but since this is still under development the datasets might only work correctly with certain versions of Thermo-Calc. The syntax of the 3rd generation datasets varies considerably and the “shorthand” notation using the THETA and LNTHETA keywords will only work with specific versions of Thermo-Calc. It is possible (with some effort) to rewrite the datasets without using those specific keywords e.g. in AlC-21He-3g.tdb.

2. Content of the datasets

The datasets are formatted rather strictly to facilitate checking, modification, and combination with other datasets. Each dataset includes a reference to the source where it was taken, a creation date and dates for any modification, or correction. There are comments about specific models used (such as ordering or associate model for the liquid), any problems noticed and if there is some mismatch between the calculated and published phase diagrams. Each parameter has a reference, pointing to the relevant literature source. Interactions for the fcc, bcc and hcp phases are included also when they are not included in the original publication. In cases where terminal solubilities are not modelled these interactions have been selected so that there is at most minor changes in the phase diagram. For modelling of ternary systems and database construction such interactions should be determined more accurately, using information from ternary systems or *ab initio* calculations using e.g. special quasirandom structures (SQS) [6]. Standard Calphad phase names (e.g. FCC_A1, SIGMA_D8B, C14_LAVES) are used. For other phases a chemical compound notation, adding a Strukturbericht symbol if available, is preferred. There is an approximately 50/50 preference for using 1 mol of atoms or 1 mol of formula units as basis for the modelling. In this collection the latter is used when possible and values based on mole of atoms are converted. This leads to problems with accurate reproduction of phase diagrams when compositions are given with insufficient accuracy. Compositions are typically given with two, three or four decimals. Two decimals is not sufficiently accurate. Three decimals is usually enough to reproduce invariant equilibria within a few K, but in rare cases, with compounds close in composition, not even four decimals is sufficient. In the collection, parameters were sometimes adjusted to compensate for these effects. Crystallographic information (prototype, Pearson symbol and space group) is added as comment for each phase. The crystallographic information is taken from the corresponding publication, Journal of Phase Equilibria and Diffusion binary reviews or the Pauling File as provided by Springer Materials [7]. In case of conflict the latter is normally considered more reliable. The crystallographic information for all phases is summarized in an Excel table provided as supplementary material.

3. Systems included

In the following all the systems included are listed with relevant references, corresponding to the datasets included. The systems are listed alphabetically, but metals are written before non-metals in any given system. The elements Ar, As, At, B, Br, C, Cl, F, H, He, I, Kr, N, Ne, O, P, Rn, S, Se, Te and Xe are considered as non-metals in this context. An enriched database file for the version 5.0 of the SGTE unary data is provided as supplementary material.

Ag-Al [8–12].
 Ag-As [13].
 Ag-Au [14].
 Ag-B [15,16].
 Ag-Ba [17].
 Ag-Be [16].
 Ag-Bi [18–21].

Ag-C [16].
 Ag-Ca [22–24].
 Ag-Cd [25,26].
 Ag-Ce [27].
 Ag-Co [28,29].
 Ag-Cr [16,30,31].
 Ag-Cu [9,19,32–35].
 Ag-Dy [36].
 Ag-Er [36,37].
 Ag-Fe [15,16].
 Ag-Ga [38,39].
 Ag-Gd [40,41].
 Ag-Ge [42–44].
 Ag-Hg [45].
 Ag-Ho [46].
 Ag-In [25,47–51].
 Ag-Ir [52].
 Ag-La [53].
 Ag-Lu [13].
 Ag-Mg [54–57].
 Ag-Mn [58].
 Ag-Mo [16].
 Ag-Na [59].
 Ag-Nb [16,60].
 Ag-Nd [40,61].
 Ag-Ni [62–64].
 Ag-O [65,66].
 Ag-Os [67].
 Ag-P [68].
 Ag-Pb [35,69–72].
 Ag-Pd [56,73–75].
 Ag-Pr [76].
 Ag-Pt [56,77].
 Ag-Pu [15].
 Ag-Rh [56,78].
 Ag-Ru [56,79].
 Ag-S [13].
 Ag-Sb [72,80,81].
 Ag-Sc [82,83].
 Ag-Se [84–86].
 Ag-Si [87,88].
 Ag-Sm [15].
 Ag-Sn [20,33,48,81,89–94].
 Ag-Sr [95].
 Ag-Tb [46].
 Ag-Te [16,70,96].
 Ag-Ti [56,97–99].
 Ag-Tl [19].
 Ag-V [16,100].
 Ag-W [16,101].
 Ag-Y [102].
 Ag-Yb [103].
 Ag-Zn [93,104–106].
 Ag-Zr [56,107].
 Al-Am [108].
 Al-As [109].
 Al-Au [110–112].
 Al-B [113–116].
 Al-Ba [117–119].
 Al-Be [120].
 Al-Bi [121–123].
 Al-C [124–129].
 Al-Ca [130–135].
 Al-Cd [136].
 Al-Ce [137–139].
 Al-Co [140–150].

Al–Cr [151–161].
 Al–Cu [9,162–165].
 Al–Dy [166–168].
 Al–Er [169,170].
 Al–Fe [171–180].
 Al–Ga [181–183].
 Al–Gd [168,184].
 Al–Ge [185–187].
 Al–H [188].
 Al–Hf [189].
 Al–Hg [190].
 Al–Ho [168,191].
 Al–In [122,183,192].
 Al–Ir [193–196].
 Al–K [197].
 Al–La [138,198–200].
 Al–Li [201–206].
 Al–Mg [130,191,207–214].
 Al–Mn [215–222].
 Al–Mo [223–228].
 Al–N [229–232].
 Al–Na [233].
 Al–Nb [234–239].
 Al–Nd [137,138,240–244].
 Al–Ni [245–254].
 Al–Np [255].
 Al–O [256–258].
 Al–P [176,259–261].
 Al–Pb [122,262,263].
 Al–Pd [264–266].
 Al–Pr [267,268].
 Al–Pt [269,270].
 Al–Pu [108,255,271].
 Al–Re [272–274].
 Al–Ru [24,275,276].
 Al–S [277].
 Al–Sb [278–280].
 Al–Sc [281–283].
 Al–Se [277].
 Al–Si [208,284–288].
 Al–Sm [289–291].
 Al–Sn [187,292,293].
 Al–Sr [294–296].
 Al–Ta [297–300].
 Al–Te [277,301].
 Al–Th [302].
 Al–Ti [303–307].
 Al–Tl [308].
 Al–U [255,309,310].
 Al–V [311–317].
 Al–W [145,318].
 Al–Y [127,319–323].
 Al–Yb [324].
 Al–Zn [181,325–328].
 Al–Zr [161,203,329–334].
 Am–Fe [335].
 Am–Ga [108].
 Am–Np [335,336].
 Am–O [337,338].
 Am–Pu [335,336,338,339].
 Am–U [335,336,340].
 Am–Zr [335].
 As–P [261].
 As–Te [341].
 Au–As [56].
 Au–B [342].
 Au–Bi [343–345].
 Au–C [56].
 Au–Ce [346,347].
 Au–Co [16,348].
 Au–Cr [56].
 Au–Cu [349].
 Au–Dy [350–352].
 Au–Er [353].
 Au–Fe [354].
 Au–Ga [355–357].
 Au–Gd [358].
 Au–Ge [359,360].
 Au–Hf [361].
 Au–Ho [362].
 Au–In [363–367].
 Au–La [353].
 Au–Lu [368].
 Au–Nd [351,369].
 Au–Ni [74,370,371].
 Au–Pb [372,373].
 Au–Pd [56].
 Au–Pr [374].
 Au–Pt [375,376].
 Au–Rh [56,377].
 Au–Ru [56,378].
 Au–Sb [379–383].
 Au–Sc [384].
 Au–Si [88,360,385].
 Au–Sn [347,348,386–390].
 Au–Tb [368,391].
 Au–Te [392,393].
 Au–Th [394].
 Au–Ti [395].
 Au–Tl [383].
 Au–Tm [362].
 Au–Yb [358].
 Au–Zn [396].
 Au–Zr [397].
 B–C [398–402].
 B–N [229,402].
 B–O [403].
 Ba–B [404].
 Ba–Bi [405].
 Ba–Ca [406].
 Ba–Co [407].
 Ba–Cu [408,409].
 Ba–Eu [410].
 Ba–Fe [407].
 Ba–Ga [411].
 Ba–Ge [117].
 Ba–Mg [412].
 Ba–Ni [413].
 Ba–O [414,415].
 Ba–Pb [416].
 Ba–Ru [17].
 Ba–Sr [417,418].
 Ba–Ti [413].
 Ba–V [419].
 Ba–Y [420].
 Ba–Yb [421].
 Be–C [422].
 Be–Cu [423].
 Be–Fe [424].
 Be–Hf [88].
 Be–Mo [425].
 Be–Pu [426].

Be-Si [422,427].
 Be-Ti [428].
 Be-V [429].
 Be-Zr [88,430].
 Bi-Ca [431].
 Bi-Ce [432,433].
 Bi-Co [434].
 Bi-Cr [435,436].
 Bi-Cs [437].
 Bi-Cu [438].
 Bi-Dy [439].
 Bi-Er [432,439,440].
 Bi-Fe [441].
 Bi-Ga [442].
 Bi-Gd [443].
 Bi-Ge [444].
 Bi-Hg [445,446].
 Bi-Ho [443].
 Bi-In [447,448].
 Bi-K [446,449,450].
 Bi-La [451].
 Bi-Li [452].
 Bi-Lu [453].
 Bi-Mg [454–456].
 Bi-Mn [457,458].
 Bi-Na [452].
 Bi-Nd [459,460].
 Bi-Ni [461–467].
 Bi-O [468].
 Bi-Pb [469–472].
 Bi-Pd [473].
 Bi-Rb [474,475].
 Bi-Rh [476].
 Bi-S [477].
 Bi-Sb [478,479].
 Bi-Se [480–483].
 Bi-Si [484].
 Bi-Sn [20,446,478,485–487].
 Bi-Sr [488].
 Bi-Tb [489].
 Bi-Te [480,490–492].
 Bi-Ti [436,493].
 Bi-Tl [494].
 Bi-Tm [437,460].
 Bi-U [457,469].
 Bi-V [436].
 Bi-Y [489].
 Bi-Yb [495].
 Bi-Zn [442,486,496–498].
 Bi-Zr [431].
 C-P [499].
 Ca-B [404].
 Ca-Br [500].
 Ca-Ce [501,502].
 Ca-Cl [500].
 Ca-Cu [503,504].
 Ca-F [500].
 Ca-Fe [505].
 Ca-Ga [411,506].
 Ca-Ge [507].
 Ca-H [508,509].
 Ca-I [500].
 Ca-In [510,511].
 Ca-Li [512].
 Ca-Mg [130,257,513–516].
 Ca-Mn [517].

Ca-Na [518].
 Ca-Ni [515].
 Ca-O [505,519–522].
 Ca-Pb [523–525].
 Ca-Ru [24].
 Ca-S [526].
 Ca-Sb [510].
 Ca-Sc [527].
 Ca-Si [135,528].
 Ca-Sn [529,530].
 Ca-Sr [531,532].
 Ca-Ti [533].
 Ca-V [419].
 Ca-Yb [531].
 Ca-Zn [534].
 Cd-B [535].
 Cd-Ce [536].
 Cd-Cu [537].
 Cd-Fe [538].
 Cd-Ga [539].
 Cd-Gd [540–542].
 Cd-Ge [543].
 Cd-Hg [544].
 Cd-In [25,539].
 Cd-Mg [545].
 Cd-Mn [538].
 Cd-Na [546].
 Cd-Pb [547–549].
 Cd-Pr [550].
 Cd-Pu [426].
 Cd-Sb [551–553].
 Cd-Se [554].
 Cd-Sn [538].
 Cd-Sr [535].
 Cd-Te [548,555,556].
 Cd-Ti [535].
 Cd-Tl [546].
 Cd-V [535].
 Cd-Y [542].
 Cd-Yb [557].
 Cd-Zn [552,558].
 Ce-B [559–561].
 Ce-C [562].
 Ce-Co [563–565].
 Ce-Cr [566].
 Ce-Cu [567,568].
 Ce-Fe [569–571].
 Ce-Ga [572,573].
 Ce-Gd [574].
 Ce-H [575].
 Ce-La [576,577].
 Ce-Mg [501,577–579].
 Ce-Mn [580,581].
 Ce-Mo [582].
 Ce-Nd [241,576].
 Ce-Ni [583–586].
 Ce-O [587,588].
 Ce-Pr [576].
 Ce-Pt [589].
 Ce-Rh [590].
 Ce-Ru [591].
 Ce-Sb [570,571,592].
 Ce-Si [593,594].
 Ce-Sn [347,595].
 Ce-Te [596].
 Ce-Ti [597].

Ce–V [598].
 Ce–W [599].
 Ce–Y [578,587,600].
 Ce–Zn [601].
 Ce–Zr [602].
 Co–B [603].
 Co–C [604,605].
 Co–Cr [606–616].
 Co–Cu [617–620].
 Co–Dy [621,622].
 Co–Er [623].
 Co–Fe [624–627].
 Co–Ga [628–630].
 Co–Gd [631,632].
 Co–Ge [16,633,634].
 Co–H [635].
 Co–Hf [636–638].
 Co–Ho [639].
 Co–In [640].
 Co–La [563,641].
 Co–Mg [642].
 Co–Mn [643].
 Co–Mo [644,645].
 Co–N [646].
 Co–Nb [235,647–649].
 Co–Nd [650].
 Co–Ni [606,651].
 Co–O [652].
 Co–Pd [75,653].
 Co–Pr [654].
 Co–Pt [614,655,656].
 Co–Re [273,606,657,658].
 Co–Ru [658].
 Co–Sb [659–662].
 Co–Sc [663].
 Co–Si [664,665].
 Co–Sm [666,667].
 Co–Sn [348,668–670].
 Co–Sr [671].
 Co–Ta [672–677].
 Co–Te [341].
 Co–Th [678].
 Co–Ti [679–684].
 Co–U [310].
 Co–V [685–689].
 Co–W [145,604,611,690–696].
 Co–Y [697–700].
 Co–Zn [701,702].
 Co–Zr [638,703–708].
 Cr–B [709–711].
 Cr–C [712–716].
 Cr–Cu [717–722].
 Cr–Fe [609,723–729].
 Cr–Ga [630,730].
 Cr–Ge [731–733].
 Cr–H [734].
 Cr–Hf [735].
 Cr–Ir [736].
 Cr–La [566,737].
 Cr–Mg [738].
 Cr–Mn [739,740].
 Cr–Mo [741].
 Cr–N [742].
 Cr–Na [59].
 Cr–Nb [743–749].
 Cr–Ni [606,608,714,750–753].
 Cr–O [754–756].
 Cr–P [757,758].
 Cr–Pb [435].
 Cr–Pd [759].
 Cr–Pt [56,614,760].
 Cr–Re [273,606,761,762].
 Cr–Ru [763,764].
 Cr–S [765,766].
 Cr–Sc [527,767].
 Cr–Si [105,768–772].
 Cr–Sn [744,773,774].
 Cr–Ta [748,775].
 Cr–Ti [735,776–781].
 Cr–U [782].
 Cr–V [778,783–785].
 Cr–W [786].
 Cr–Y [566,697].
 Cr–Zn [787–789].
 Cr–Zr [717,746,773,790–794].
 Cs–In [795].
 Cs–K [796,797].
 Cs–Mo [798].
 Cs–Na [796,797].
 Cs–O [799].
 Cs–Rb [796,797].
 Cs–Te [800].
 Cu–As [801,802].
 Cu–B [803,804].
 Cu–C [64,805].
 Cu–Dy [806,807].
 Cu–Er [808].
 Cu–Eu [809].
 Cu–Fe [810–816].
 Cu–Ga [817,818].
 Cu–Gd [819].
 Cu–Ge [820].
 Cu–H [821,822].
 Cu–Hf [823–825].
 Cu–Hg [45].
 Cu–Ho [826].
 Cu–In [827–830].
 Cu–Ir [16,831].
 Cu–La [832–834].
 Cu–Li [835–837].
 Cu–Mg [838–841].
 Cu–Mn [842–844].
 Cu–Mo [845].
 Cu–N [846].
 Cu–Na [59].
 Cu–Nb [721].
 Cu–Nd [568,847,848].
 Cu–Ni [816,849–851].
 Cu–O [65,821,852–854].
 Cu–P [855–858].
 Cu–Pb [35,859–861].
 Cu–Pd [822,862].
 Cu–Pr [568].
 Cu–Pt [863].
 Cu–Pu [864].
 Cu–Rh [865].
 Cu–S [866,867].
 Cu–Sb [868–870].
 Cu–Sc [83,281].
 Cu–Se [868,871].
 Cu–Si [872–879].
 Cu–Sm [568].

Cu–Sn [48,164,855,880–884].
 Cu–Sr [504].
 Cu–Ta [885].
 Cu–Te [886–888].
 Cu–Th [889].
 Cu–Ti [88,882,890–893].
 Cu–Tl [894].
 Cu–Tm [804].
 Cu–U [895].
 Cu–V [721,896].
 Cu–W [101,897,898].
 Cu–Y [899].
 Cu–Yb [809].
 Cu–Zn [900–905].
 Cu–Zr [32,88,107,704,906–912].
 Dy–B [559].
 Dy–C [913].
 Dy–Er [914].
 Dy–Fe [915–917].
 Dy–Ge [918].
 Dy–H [919].
 Dy–Ho [914].
 Dy–Mg [168,920].
 Dy–Mn [921,922].
 Dy–Ni [923].
 Dy–O [924,925].
 Dy–Pb [926].
 Dy–Pd [927].
 Dy–Ru [928].
 Dy–Sb [929].
 Dy–Si [913].
 Dy–Sn [930].
 Dy–Tb [917].
 Dy–Te [931].
 Dy–W [599].
 Dy–Zn [932].
 Dy–Zr [167].
 Er–B [113,933,934].
 Er–Fe [915,934–936].
 Er–Ge [937].
 Er–H [938].
 Er–Ho [914].
 Er–Lu [939].
 Er–Mg [170].
 Er–Mn [580,654].
 Er–O [924,925].
 Er–Pd [940].
 Er–Sb [941].
 Er–Sc [942].
 Er–Tb [914].
 Er–Te [440].
 Er–Ti [597].
 Er–V [623].
 Er–W [599].
 Er–Y [37].
 Er–Zn [932].
 Er–Zr [943].
 Eu–Ga [944].
 Eu–In [945].
 Eu–Mg [946].
 Eu–Ni [947].
 Eu–O [925].
 Eu–Pb [948].
 Eu–Pd [949].
 Eu–Sn [950].
 Eu–Te [596,951].

Eu–W [599].
 Fe–As [952,953].
 Fe–B [711,954–959].
 Fe–C [215,398,805,960–965].
 Fe–Ga [271].
 Fe–Gd [632,966–969].
 Fe–H [970].
 Fe–Hf [971].
 Fe–Ho [972,973].
 Fe–In [974].
 Fe–La [569,975].
 Fe–Lu [915,976].
 Fe–Mg [977].
 Fe–Mn [960,978–983].
 Fe–Mo [954,984–986].
 Fe–N [742,987–990].
 Fe–Nb [724,956,991–1004].
 Fe–Nd [1005–1007].
 Fe–Ni [250,723,727,960,1008–1013].
 Fe–Np [335].
 Fe–O [522,975,1014–1018].
 Fe–Os [1019].
 Fe–P [499,757,1020,1021].
 Fe–Pb [1022,1023].
 Fe–Pd [75].
 Fe–Pr [1006,1024].
 Fe–Pt [1025].
 Fe–Pu [271,1026].
 Fe–Rh [1027].
 Fe–Ru [764].
 Fe–S [866,867,1028].
 Fe–Sb [441,1029,1030].
 Fe–Sc [83,384,663].
 Fe–Si [1008,1031–1035].
 Fe–Sm [966,970].
 Fe–Sn [1036–1038].
 Fe–Sr [1039].
 Fe–Ta [1040–1042].
 Fe–Tb [915–917].
 Fe–Te [1043,1044].
 Fe–Th [678].
 Fe–Ti [303,955,978,1045–1055].
 Fe–Tm [972,976].
 Fe–U [335,1014,1056,1057].
 Fe–V [992,1048,1058–1060].
 Fe–W [1061,1062].
 Fe–Y [915,1063–1066].
 Fe–Zn [1067–1071].
 Fe–Zr [790,993,1036,1072–1080].
 Ga–As [1081].
 Ga–B [411].
 Ga–Gd [572,1082,1083].
 Ga–Ge [187].
 Ga–Hg [1084,1085].
 Ga–Ho [1086].
 Ga–In [183,1087,1088].
 Ga–La [1089,1090].
 Ga–Li [1091].
 Ga–Lu [1092].
 Ga–Mg [371,1093–1095].
 Ga–N [1096,1097].
 Ga–Na [1098].
 Ga–Nb [1082].
 Ga–Ni [630,1099].
 Ga–O [1100].
 Ga–P [261].

Ga–Pb [1101,1102].
 Ga–Pd [630].
 Ga–Pr [1089].
 Ga–Pt [1103,1104].
 Ga–Pu [271].
 Ga–Sb [261,1105].
 Ga–Sc [1106].
 Ga–Se [1107].
 Ga–Si [371,1108].
 Ga–Sn [187,1109].
 Ga–Sr [411].
 Ga–Tb [1106].
 Ga–Te [887,1110,1111].
 Ga–Ti [1112,1113].
 Ga–Tl [1114].
 Ga–U [1115,1116].
 Ga–V [1117].
 Ga–Y [1118].
 Ga–Zn [181,1119].
 Ga–Zr [1120].
 Gd–B [1121].
 Gd–Ge [1122].
 Gd–H [1123].
 Gd–Li [1124].
 Gd–Mg [168,1125–1128].
 Gd–Mn [1129,1130].
 Gd–Mo [968,1131].
 Gd–Nb [1082].
 Gd–Ni [1132,1133].
 Gd–O [924,925,1134].
 Gd–Pb [926,1135].
 Gd–Pd [1136].
 Gd–Ru [928].
 Gd–Sb [1137].
 Gd–Sc [767,1138].
 Gd–Si [1139].
 Gd–Sm [1140].
 Gd–Sr [1125].
 Gd–Te [931].
 Gd–Ti [1141].
 Gd–Tl [1142].
 Gd–W [599].
 Gd–Y [574,1128].
 Gd–Zn [1143].
 Gd–Zr [184,1131,1144,1145].
 Ge–As [109,1146,1147].
 Ge–C [1148].
 Ge–Hf [1149].
 Ge–In [1150].
 Ge–K [1151].
 Ge–Li [1152,1153].
 Ge–Lu [918].
 Ge–Mg [185,1154].
 Ge–Mn [1155,1156].
 Ge–Na [1151].
 Ge–Nb [1157,1158].
 Ge–Ni [1159,1160].
 Ge–O [1161,1162].
 Ge–P [1146,1163].
 Ge–Pb [1150].
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 Ge–S [1146].
 Ge–Sb [1150,1165–1167].
 Ge–Sc [1168].
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 Ge–Sr [1172].
 Ge–Ta [1173].
 Ge–Te [1174].
 Ge–Ti [1175,1176].
 Ge–Tl [1150].
 Ge–Tm [1177].
 Ge–V [1178,1179].
 Ge–Y [1180].
 Ge–Yb [1181,1182].
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 Ge–Zr [1183].
 Hf–B [1184–1187].
 Hf–C [1188,1189].
 Hf–Ir [1190].
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 Hf–Mo [1193,1194].
 Hf–N [1195].
 Hf–Nb [1196,1197].
 Hf–Ni [1185,1198–1201].
 Hf–O [1202,1203].
 Hf–Os [1204].
 Hf–Pt [1205].
 Hf–Ru [275].
 Hf–Si [1206–1208].
 Hf–Sn [1209,1210].
 Hf–Ta [1211].
 Hf–Ti [1187].
 Hf–V [1212].
 Hf–W [1194,1213].
 Hf–Zr [1214–1216].
 Hg–In [1217,1218].
 Hg–Mg [1084].
 Hg–Pb [445].
 Hg–Sn [1219].
 Hg–Te [556,1220].
 Hg–Tl [1221].
 Hg–Zn [1218].
 Ho–B [933].
 Ho–Mg [168,191].
 Ho–Mn [921,1222].
 Ho–Mo [826].
 Ho–Ni [583,1223].
 Ho–O [924,925].
 Ho–Pd [1224].
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 Ho–Te [931,1228].
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 In–Ir [1229].
 In–La [1230].
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 In–N [1234].
 In–Na [511].
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 In–O [1237].
 In–P [261].
 In–Pb [373,472,1238].
 In–Pd [1239].
 In–Pt [1240].

In-S [1241].
 In-Sb [261,381,660,1105,1236].
 In-Sc [1242].
 In-Se [1243].
 In-Si [1244].
 In-Sn [51,487,1238,1245,1246].
 In-Sr [488].
 In-Te [301].
 In-V [1247].
 In-Y [1242].
 In-Yb [945].
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 Ir-C [16].
 Ir-Mo [1249].
 Ir-Ni [16].
 Ir-Pd [56].
 Ir-Pt [1250].
 Ir-Rh [1250].
 Ir-Ru [1250].
 Ir-Ti [1251].
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 La-W [599].
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 Li-N [1293].
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 Li-Sr [1304].
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 Li-V [100].
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 Lu-B [559].
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 Lu-O [925].
 Lu-Pb [948,1307].
 Lu-Sb [453].
 Lu-W [599].
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 Mg-Cl [1253].
 Mg-H [1311,1312].
 Mg-Mn [1313,1314].
 Mg-N [1315].
 Mg-Na [233].
 Mg-Nb [1315].
 Mg-Nd [578,1316–1320].
 Mg-Ni [515,1321,1322].
 Mg-O [256,521,1323].
 Mg-Pb [454,1324].
 Mg-Pr [1325–1327].
 Mg-Pu [864].
 Mg-Ru [1328].
 Mg-S [526].
 Mg-Sb [279,1329,1330].
 Mg-Sc [1331,1332].
 Mg-Si [286,1333–1338].
 Mg-Sm [290,1140,1339].
 Mg-Sn [74,1329,1340–1342].
 Mg-Sr [1343,1344].
 Mg-Tb [1345].
 Mg-Ti [1346].
 Mg-Tl [1347].
 Mg-Tm [1348].
 Mg-V [83,1315].
 Mg-Y [1128,1349–1352].
 Mg-Yb [1345].
 Mg-Zn [1032,1317,1353–1356].
 Mg-Zr [1357–1359].
 Mn-B [1360].
 Mn-C [215,981,1361–1363].
 Mn-H [1364,1365].
 Mn-Mo [1366].
 Mn-N [1367].
 Mn-Nb [998,1368].
 Mn-Nd [1369].
 Mn-Ni [1370–1373].
 Mn-O [1374,1375].
 Mn-P [1373,1376,1377].
 Mn-Pb [1378].
 Mn-Pd [1379].
 Mn-Pr [580,922].
 Mn-S [1380–1383].
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 Mn-Sc [1385].
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 Mn-Te [1392].

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 Mn–V [1398].
 Mn–W [1399,1400].
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 Mn–Zn [1353,1402–1404].
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 Mo–C [1411].
 Mo–N [742,1412].
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 Mo–Ni [1416–1420].
 Mo–O [1421,1422].
 Mo–P [1423].
 Mo–Pd [759].
 Mo–Pt [1424].
 Mo–Re [1414,1425–1428].
 Mo–Ru [275,1429].
 Mo–S [1430].
 Mo–Sc [767].
 Mo–Si [228,1431–1435].
 Mo–Ta [1413,1415,1419,1436–1438].
 Mo–Th [889].
 Mo–Ti [1412,1439–1445].
 Mo–U [309,782,1421].
 Mo–V [1446–1448].
 Mo–W [1413,1449].
 Mo–Y [582].
 Mo–Zr [1193,1439,1450,1451].
 Na–F [1254].
 Na–H [1289,1452].
 Na–O [1453].
 Na–Rb [796,797].
 Na–Sb [1255,1454].
 Na–Si [59].
 Na–Sn [1454].
 Na–Sr [1304].
 Na–Te [1256].
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 Na–V [100].
 Na–Zn [354].
 Na–Zr [1214].
 Nb–B [956,1456–1459].
 Nb–C [745,997,1001,1460,1461].
 Nb–F [1462].
 Nb–H [1463,1464].
 Nb–N [997,1465,1466].
 Nb–Ni [648,1467–1471].
 Nb–O [1472,1473].
 Nb–P [1474].
 Nb–Re [1414,1475].
 Nb–Sb [1476].
 Nb–Si [769,1207,1433,1477–1482].
 Nb–Sn [744,1483,1484].
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 Nb–Ti [1440,1441,1485–1487].
 Nb–U [1397].
 Nb–V [1487,1488].
 Nb–W [1413,1460].
 Nb–Y [1489].
 Nb–Zn [1490].
 Nb–Zr [773,1491].
 Nd–B [342,1007,1492,1493].
 Nd–C [1494].
 Nd–H [1495].
 Nd–Ni [583,1495–1497].
 Nd–O [925,1498].
 Nd–Pr [1499].
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 Nd–Ti [1504].
 Nd–U [1498].
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 Ni–As [801].
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 Ni–C [714,1513,1514].
 Ni–H [1257,1515].
 Ni–N [1516].
 Ni–O [756,1016].
 Ni–Os [1517].
 Ni–P [1518–1520].
 Ni–Pb [861,1521–1523].
 Ni–Pd [75].
 Ni–Pr [1524,1525].
 Ni–Pt [1526].
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 Ni–Rh [1529].
 Ni–Ru [50,1530,1531].
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 Ni–Sc [1535–1537].
 Ni–Se [1538].
 Ni–Si [1008,1512,1539–1545].
 Ni–Sm [1546,1547].
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 Ni–Y [320,697,1584].
 Ni–Yb [947,1585].
 Ni–Zn [903,1586–1588].
 Ni–Zr [88,1185,1589–1593].
 Np–O [1594].
 Np–Pu [335].
 Np–U [335,1595].
 Np–Zr [335,1595–1597].
 Os–C [1598].
 Os–Pt [1599].
 Os–Si [1600].
 Pb–As [1601].
 Pb–C [1602].
 Pb–O [1603].
 Pb–Pd [1604,1605].
 Pb–Pt [1606].
 Pb–Pu [1607].
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 Pb–Sc [926].
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 Pb–Tl [1616].
 Pb–U [469,1607].
 Pb–Y [1617].
 Pb–Yb [1618].
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 Pu–C [1660,1661].
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 Pu–Th [1664].
 Pu–U [340,1662,1665,1666].
 Pu–Zn [1274].
 Pu–Zr [1665,1666].
 Re–B [1667].
 Re–Ru [1668,1669].
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 Re–Ti [1673].
 Re–U [782].
 Re–W [1671,1674].
 Re–Y [1675].
 Re–Zr [1676,1677].
 Rh–C [1598].
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 Ru–Zr [764,1688].
 S–Se [1689].
 Sb–As [261].
 Sb–P [261].
 Sb–S [1241,1690].
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 Sb–Si [380,1693].
 Sb–Sm [1694,1695].
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 Sb–Tb [1275].
 Sb–Te [1701,1702].
 Sb–Tm [941].
 Sb–Y [1703].
 Sb–Yb [1226,1704].
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 Sc–B [82,1709].
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 Sc–Si [83,1710].
 Sc–Sn [1711].
 Sc–Sr [1304].
 Sc–Th [82].
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 Sc–Y [102,1138].
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 Sc–Zr [83,767,1214,1713].
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 Si–As [1147].
 Si–B [1511,1512,1717,1718].
 Si–C [1035,1286,1623,1719].
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 Si–Sr [1730–1732].
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 Si–V [1740–1743].
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 Sm–W [599].
 Sm–Zn [1505,1751].
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 Sn–O [1752].

Sn-P [855,856,858].
 Sn-S [1753].
 Sn-Se [1754].
 Sn-Sr [1755,1756].
 Sn-Ta [1757].
 Sn-Tb [1502].
 Sn-Te [1758].
 Sn-Ti [882,1759–1762].
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 Sn-Y [1210,1767].
 Sn-Yb [557,1768].
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 Zr-O [1014,1202,1840,1872,1893–1895].
 Zr-Te [596].

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.calphad.2025.102833>.

Data availability

The data that supports the findings of this study are available in the supplementary material of this article.

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